HDOC DAY – 6

Name-Reyansh Pandit

SAP ID-590033602

Batch-16

Date-29 August 2026

Question 1:

Write a C program to find the area and circumference in meters of a circle when its diameter is given in centimetres.

Understand the problem statement and identify the required input, output, and formula.

1. Understanding the Problem Statement

Input

The program takes:

- Diameter of the circle in centimetres (cm).

Required Output

The program should calculate:

- Area of the circle in square meters (m^2)
- Circumference of the circle in meters (m)

Unit Conversion

Since the diameter is given in centimeters but the output is required in meters:

1 meter = 100 centimeters

Therefore:

Diameter in meters = Diameter in centimeters / 100

Formula Used

Let:

- d = diameter in meters
- r = radius in meters

Radius:

r = d / 2

Area:

Area = π × r × r

Circumference:

Circumference = π × d

For the program, we can use:

π = 3.14159

2. Standard Test Case Table

Test Case	Input Diameter (cm)	Expected Output	Actual Output	Result
1	100	Area = 0.785398 m^2, Circumference = 3.141590 m	Area = 0.785398 m^2, Circumference = 3.141590 m	Pass
2	200	Area = 3.141590 m^2, Circumference = 6.283180 m	Area = 3.141590 m^2, Circumference = 6.283180 m	Pass
3	50	Area = 0.196349 m^2, Circumference = 1.570795 m	Area = 0.196349 m^2, Circumference = 1.570795 m	Pass
4	10	Area = 0.007854 m^2, Circumference = 0.314159 m	Area = 0.007854 m^2, Circumference = 0.314159 m	Pass
5	250	Area = 4.908734 m^2, Circumference = 7.853975 m	Area = 4.908734 m^2, Circumference = 7.853975 m	Pass

3. Algorithm

Algorithm: To Calculate Area and Circumference of a Circle

1. Start.
2. Declare variables for diameter in centimeters, diameter in meters, radius, area, and circumference.
3. Input the diameter of the circle in centimeters.
4. Convert the diameter from centimeters to meters by dividing it by 100.
5. Calculate the radius by dividing the diameter in meters by 2.
6. Calculate the area using the formula π × radius × radius.
7. Calculate the circumference using the formula π × diameter.

8. Display the area in square meters.
9. Display the circumference in meters.
10. Stop.

---

4. Evaluation of the Algorithm

The algorithm correctly solves the given problem because it first converts the input diameter from centimeters into meters. This is necessary because the required output must be expressed in meters. After conversion, the radius is calculated by dividing the diameter by two. The area is then calculated using the formula πr^2, while the circumference is calculated using πd. The algorithm is simple, sequential, and efficient because it performs a fixed number of operations without using loops or complex conditions.

5. C Program

```
#include <stdio.h>


int main()

{

   float diameter_cm, diameter_m, radius;

   float area, circumference;

   const float PI = 3.14159;


   printf("Enter the diameter of the circle in centimeters: ");

   scanf("%f", &diameter_cm);


   diameter_m = diameter_cm / 100;


   radius = diameter_m / 2;


   area = PI * radius * radius;


   circumference = PI * diameter_m;
```

```
    printf("\nDiameter in meters = %.4f m", diameter_m);
    printf("\nArea of the circle = %.6f square meters", area);
    printf("\nCircumference of the circle = %.6f meters\n", circumference);


    return 0;
}
```

6. Explanation of the Program

Header File

```
#include <stdio.h>
```

The stdio.h header file is included to use standard input and output functions such as printf() and scanf().

Variable Declaration

The program declares variables to store:

- Diameter in centimeters
- Diameter in meters
- Radius
- Area
- Circumference

The value of π is stored as a constant:

const float PI = 3.14159;

Input

The user enters the diameter in centimeters.

Conversion

The diameter is converted into meters:

diameter_m = diameter_cm / 100

Radius Calculation

The radius is half of the diameter:

radius = diameter_m / 2

Area Calculation

The area is calculated using:

Area = π × radius × radius

Circumference Calculation

The circumference is calculated using:

Circumference = π × diameter

Finally, the calculated values are displayed.

---

7. Compilation

Save the program with the filename:

circle.c

Compile it using CC:

cc circle.c -

If there are no errors, execute the program using:

./a.out

---

8. Sample Execution

Test Case

Enter the diameter of the circle in centimeters: 100

Diameter in meters = 1.0000 m

Area of the circle = 0.785398 square meters

Circumference of the circle = 3.141590 meters

---

9. Evaluation Against Test Cases

The program was tested using different values of diameter, including small, medium, and large values. In each test case, the diameter was correctly converted from centimeters to meters before performing the calculations. The area and circumference produced by the program matched the expected values.

Therefore, all the prepared test cases were successfully passed.

---

10. Result

Thus, the C program to calculate the area and circumference of a circle in meters, when the diameter is given in centimeters, was successfully written, compiled, and executed. The program was tested with multiple test cases and produced the expected results.

Question 2:

Write a C program to find the segment area and coloring cost of a circular segment when its sector area, triangle area, and coloring rate per square centimetre are given.

1. Understanding the Problem Statement

Input

The program takes:

- Sector area in square centimeters (cm^2)
- Triangle area in square centimeters (cm^2)
- Coloring rate per square centimeter

Required Output

The program should calculate:

- Area of the circular segment in square centimeters (cm^2)
- Total coloring cost in Rupees (₹)

Formula Used

Let:

- sector_area = Area of the sector
- triangle_area = Area of the triangle
- segment_area = Area of the circular segment
- coloring_rate = Cost per square centimeter

Segment Area

Segment Area = Sector Area − Triangle Area

Coloring Cost

Coloring Cost = Segment Area × Coloring Rate

2. Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	Sector Area = 100 cm^2, Triangle Area = 60 cm^2, Rate = ₹5/cm^2	Segment Area = 40 cm^2, Cost = ₹200	Segment Area = 40 cm^2, Cost = ₹200	Pass
2	Sector Area = 200 cm^2, Triangle Area = 120 cm^2, Rate = ₹10/cm^2	Segment Area = 80 cm^2, Cost = ₹800	Segment Area = 80 cm^2, Cost = ₹800	Pass
3	Sector Area = 50 cm^2, Triangle Area = 30 cm^2, Rate = ₹8/cm^2	Segment Area = 20 cm^2, Cost = ₹160	Segment Area = 20 cm^2, Cost = ₹160	Pass
4	Sector Area = 150 cm^2, Triangle Area = 100 cm^2, Rate = ₹12/cm^2	Segment Area = 50 cm^2, Cost = ₹600	Segment Area = 50 cm^2, Cost = ₹600	Pass
5	Sector Area = 500 cm^2, Triangle Area = 350 cm^2, Rate = ₹4/cm^2	Segment Area = 150 cm^2, Cost = ₹600	Segment Area = 150 cm^2, Cost = ₹600	Pass

3. Algorithm

Algorithm: To Calculate Segment Area and Coloring Cost of a Circular Segment

1. Start.
2. Declare variables for sector area, triangle area, segment area, coloring rate, and coloring cost.
3. Input the sector area in square centimeters.
4. Input the triangle area in square centimeters.
5. Input the coloring rate per square centimeter.
6. Calculate the segment area by subtracting the triangle area from the sector area.

7. Calculate the coloring cost by multiplying the segment area by the coloring rate.
8. Display the segment area in square centimeters.
9. Display the coloring cost in Rupees.
10. Stop.

---

4. Evaluation of the Algorithm

The algorithm correctly solves the given problem because it first calculates the area of the circular segment by subtracting the triangle area from the sector area. This is based on the mathematical relationship between a circular sector, the triangle formed inside it, and the remaining circular segment.

After calculating the segment area, the algorithm calculates the total coloring cost by multiplying the segment area by the given coloring rate per square centimeter.

The algorithm is simple, sequential, and efficient because it performs a fixed number of arithmetic operations without using loops or complex conditions

5. C Program

```
#include <stdio.h>


int main()

{

   float sector_area, triangle_area, segment_area;

   float coloring_rate, coloring_cost;


   printf("Enter the sector area in square centimeters: ");

   scanf("%f", &sector_area);


   printf("Enter the triangle area in square centimeters: ");

   scanf("%f", &triangle_area);


   printf("Enter the coloring rate per square centimeter: ");

   scanf("%f", &coloring_rate);
```

```
    segment_area = sector_area - triangle_area;

    coloring_cost = segment_area * coloring_rate;

    printf("\nSegment Area = %.2f square centimeters", segment_area);
    printf("\nColoring Cost = Rs. %.2f\n", coloring_cost);

    return 0;
}
```

6. Explanation of the Program

Header File

#include <stdio.h>

The stdio.h header file is included to use standard input and output functions such as printf() and scanf().

Variable Declaration

The program declares variables to store:

- Sector area
- Triangle area
- Segment area
- Coloring rate
- Coloring cost

The float data type is used because area and cost values may contain decimal numbers.

Input

The user enters:

- The area of the sector
- The area of the triangle
- The coloring rate per square centimeter

Segment Area Calculation

The area of the circular segment is calculated using:

Segment Area = Sector Area - Triangle Area

Coloring Cost Calculation

The total coloring cost is calculated using:

Coloring Cost = Segment Area × Coloring Rate

Finally, the calculated segment area and coloring cost are displayed.

---

7. Compilation

Save the program with the filename:

segment.c

Compile it using CC:

cc segment.c

If there are no errors, execute the program using:

./a.out

---

8. Sample Execution

Test Case

Enter the sector area in square centimeters: 100

Enter the triangle area in square centimeters: 60

Enter the coloring rate per square centimeter: 5

Segment Area = 40.00 square centimeters

Coloring Cost = Rs. 200.00

---

9. Evaluation Against Test Cases

The program was tested using different values of sector area, triangle area, and coloring rate. In each test case, the segment area was correctly calculated by subtracting the triangle area from the sector area. The coloring cost was then calculated by multiplying the segment area by the given coloring rate.

The segment area and coloring cost produced by the program matched the expected values.

Therefore, all the prepared test cases were successfully passed.

10. Result

Thus, the C program to calculate the segment area and coloring cost of a circular segment, when the sector area, triangle area, and coloring rate per square centimeter are given, was successfully written, compiled, and executed. The program was tested with multiple test cases and produced the expected results.